EXPERIMENT : 8

AIM:

Conduct Z test for the Data Given using Python.

OBJECTIVE:

To test whether the average weight of a species of birds differs from 150 grams.

PROCEDURE:

1. Null Hypothesis (H0): The average weight of the birds is 150 grams.
2. Alternative Hypothesis (H1): The average weight of the birds is not 150 grams.
3. Sample: Measure the weights of 30 birds randomly selected from the population.
4. Z-Test: Conduct a Z-test to compare the sample mean to 150 grams.
5. Decision Rule: Use a significance level of α = 0.05.

CODE:

```
import numpy as np
import scipy.stats as stats

# Define the sample data (hypothetical weights in grams)
sample_data = np.array([
    158, 162, 160, 159, 161,
    163, 157, 160, 162, 159,
    161, 158, 160, 163, 159,
    162, 161, 158, 160, 159,
    163, 160, 161, 158, 162,
    159, 160, 161, 163
])
```

```
# Population mean under the null hypothesis
population_mean = 160

# Calculate sample statistics
sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1)

# Number of observations
n = len(sample_data)

# Calculate the Z-statistic
z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

# Calculate the two-tailed p-value
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))

# Print results
print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")

# Decision based on the significance level
alpha = 0.05

if p_value < alpha:
    print("Reject the null hypothesis: The average weight is significantly different from 160 grams.")
else:
    print("Fail to reject the null hypothesis: There is no significant difference in average weight from 160 grams.")
```

OUTPUT:

```
Sample Mean: 160.31
Z-Statistic: 0.9637
P-Value: 0.3352
Fail to reject the null hypothesis: There is no significant
difference in average weight from 160 grams.
```

RESULT:

The Z-test was successfully conducted using Python to determine whether the average weight of the bird species differs significantly from 150 grams. The hypothesis was tested using the calculated Z-statistic and p-value, and the conclusion was drawn based on the significance level.